# CYBERSIGHT

## Comprehensive Technical Architecture & Project Implementation Report

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|---|---|
| **Project Title:** | CYBERSIGHT — Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance |
| **Target Agency / Ecosystem:** | Indian Cyber Crime Coordination Centre (I4C), Ministry of Home Affairs (MHA), State Law Enforcement Agencies (LEAs), Citizen Financial Cyber Fraud Reporting & Management System (CFCFRMS / 1930) |
| **Classification:** | Operational Law Enforcement & Financial Intelligence Framework |
| **Report Generated:** | September 2026 |

Smart India Hackathon 2026 — Technical Submission

# Contents

# Executive Summary

CYBERSIGHT is an end-to-end, proactive cybercrime mitigation framework engineered to solve the critical "Golden Hour" bottleneck in financial cyber fraud. Presently, the National Cybercrime Reporting Portal (NCRP) receives over 8,000 complaints daily. Criminal syndicates rapidly funnel stolen capital across Layer 1, Layer 2, and Layer 3 mule accounts before physically extracting paper currency via automated teller machines (ATMs), micro-ATMs, and Customer Service Points (CSPs).

Traditional law enforcement responses are **retrospective** (freezes occur hours or days after physical cashout). CYBERSIGHT shifts this paradigm from **reactive freezing to proactive spatial interception** by:

1. Ingesting live transaction and complaint streams.
2. Modeling spatial-temporal syndicate tendencies via 2D Gaussian Kernel Density Estimation (KDE) and geodesic distance matrices.
3. Constructing multi-hop Directed Acyclic Graphs (DAGs) of mule money trails using NetworkX.
4. Computing physical ATM vulnerability scores ($V_{\text{ATM}}$) and dynamic countdown ETAs ($\tau_{\text{ETA}}$).
5. Triggering simultaneous automated interventions: geo-fenced PCR police vehicle dispatch routes and 1-click legal bank account liens under Section 94 BNSS / Section 91 CrPC.

# Complete Technical Stack & Dependencies

## Backend Core Stack

| Technology / Library | Version | Operational Function in CYBERSIGHT |
|---|---|---|
| **Python** | 3.11+ / 3.14 | Primary runtime engine for analytics, mathematics, and API servers |
| **FastAPI** | 0.115+ | High-performance asynchronous REST API framework handling 16 endpoints |
| **Uvicorn** | 0.30+ | ASGI lightning-fast web server powering HTTP and WebSocket protocols |
| **Pydantic** | 2.x | Strict data validation, schema enforcement, and JSON serialization |
| **NetworkX** | 3.2+ | Directed Graph (DAG) construction, topological sorting, and money flow traversal |
| **NumPy** | 1.26+ | Vectorized matrix operations, Haversine computations, and spatial KDE calculations |
| **SQLite 3 / PostgreSQL** | 3.x / 14+ | Relational persistence of complaints, layer hops, ATMs, and audit trails |

## Frontend Core Stack

| Technology / Library | Version | Operational Function in CYBERSIGHT |
|---|---|---|
| **React** | 19.0.0 | Declarative component UI engine powering reactive government dashboards |
| **Vite** | 6.x | Next-generation bundler and HMR dev server with proxy routing |
| **TailwindCSS** | 4.0 | Modern utility-first CSS styling strictly implementing MHA light mode |
| **Leaflet & React-Leaflet** | 1.9.4 | Tactical GIS mapping layer rendering geolocated ATMs and radar heatmaps |
| **Lucide React** | 0.475+ | Standardized iconography across dispatch, bank, radar, and threat badges |
| **Web Audio API** | Native | Zero-external-asset dynamic sine/gain oscillator for control-room audio chimes |

### Knowledge Graph & Static Analysis

| Tool | Version | Operational Function in CYBERSIGHT |
|---|---|---|
| **Graphify** | 0.9.55 | AST structural code extraction, Louvain community clustering, and graph audit |

## Algorithmic Formulations & Mathematical Models

The intelligence core of CYBERSIGHT is implemented inside `backend/models/predictive_engine.py`. It fuses spatial heuristics, temporal velocity, and physical vulnerability metrics.

### Geodesic Proximity: Haversine Formulation

To calculate the physical distance between an ATM coordinate $(\phi_1, \lambda_1)$ and a reported cybercrime corridor centroid $(\phi_2, \lambda_2)$:

$$\Delta\phi = \phi_2 - \phi_1, \quad \Delta\lambda = \lambda_2 - \lambda_1$$

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)$$

$$c = 2 \cdot \text{atan2}\left(\sqrt{a}, \sqrt{1-a}\right), \quad d = R \cdot c \quad (R = 6371.0 \text{ km})$$

### Spatial Probability Density: 2D Gaussian Kernel Density Estimation (KDE)

The likelihood $P_{\text{spatial}}(x)$ of an ATM at position $x = (\text{lat}, \text{lng})$ being chosen by a cash-out syndicate operating within an active corridor of historical withdrawal coordinates $\{x_1, x_2, \ldots, x_n\}$ is modeled as:

$$P_{\text{spatial}}(x) = \frac{1}{n \cdot 2\pi h^2} \sum_{i=1}^{n} \exp\left(-\frac{\|x - x_i\|^2}{2h^2}\right)$$

*Where $h$ represents the bandwidth smoothing parameter (calibrated to $0.08^\circ \approx 8.8$ km based on urban mule dispersal patterns).*

### Physical ATM Vulnerability Index ($V_{\text{ATM}}$)

Not all ATMs are equally attractive to cash-out mules. CYBERSIGHT computes a vulnerability score $V_{\text{ATM}} \in [0.10, 0.98]$ weighted by physical security indicators:

$$V_{\text{ATM}} = w_{\text{type}} \cdot T + w_{\text{cctv}} \cdot (1 - C) + w_{\text{loc}} \cdot L + w_{\text{limit}} \cdot \left(\frac{\text{CashLimit}}{100000}\right)$$

- **Terminal Type Weight ($T$):** Standalone Off-Site (1.0) vs. High-Street Kiosk (0.75) vs. On-Site Bank Branch (0.35).
- **CCTV Coverage ($C$):** None (0.0 $\implies 1 - C = 1.0$), Single Static (0.5), 24/7 AI-monitored (1.0 $\implies 1 - C = 0.0$).
- **Location Profile ($L$):** Transit Hub / Interstate Border (0.95) vs. Industrial Outskirt (0.80) vs. Regulated Market (0.40).

### Golden-Hour Cash-Out ETA Regressor ($\tau_{\text{ETA}}$)

The estimated countdown time until physical cash extraction at the target terminal is calculated as a function of reported transaction velocity and mule transit friction:

$$\tau_{\text{ETA}} = \max\left(12,\ \tau_{\text{base}} - \left(\frac{\text{Amount}}{150000} \times 8\right) + \left(\frac{d_{\text{ATM}}}{v_{\text{mule}}} \times 60\right) - (\text{HopCount} \times 4)\right)$$

*If $\tau_{ETA} \leq 30$ minutes, the prediction is promoted to **CRITICAL RISK** with immediate automated PCR sirens and dispatch triggers.*

### Composite Confidence Score ($S_{\text{confidence}}$)

$$S_{\text{confidence}} = \left(0.40 \cdot P_{\text{spatial}} + 0.35 \cdot V_{\text{ATM}} + 0.25 \cdot \text{CorridorAffinity}\right) \times 100\%$$

## Multi-Layer Money Trail Network Forensics

Implemented in `backend/services/graph_forensics.py` using **NetworkX Directed Acyclic Graphs (DAG)**:

```
[Victim Account] (Loss: Rs 4,80,000)
       |
       v  UTR: 402918401928 (Layer 1 Mule - Current A/c)
[Layer 1 Node: SBI Khari Baoli]
       |
       +--------------+--------------+
       v (Rs 1,80,000) v (Rs 1,50,000) v (Rs 1,50,000)
 [L2 Mule: ICICI] [L2 Mule: HDFC]  [L2 Mule: Canara]
       |
```

```
      v  Immediate IMPS Spurt (Layer 3 Mule / Pre-Paid Card)
 [Layer 3 Mule: Axis Bank Nuh]
       |
      v  ATM Card Cloned / Micro-ATM Cash-Out Kiosk
 [TARGET] FORECASTED TERMINAL: PNB Off-Site Kiosk, Punhana Pin: 122508
   ETA: 24 Mins | Confidence: 94% | Nearest PCR: PCR-NUH-14 (3.2 km away)
```

1. **Topological Sorting:** Verifies directional flow of stolen capital and identifies splitting/layering hops.
2. **Node Centrality Scoring:** Identifies key laundering aggregators (mule hubs receiving transactions from multiple victims).
3. **Automated Interception Nodes:** Identifies which intermediate bank branch account can be frozen to paralyze multiple downstream cash-out attempts simultaneously.

## System Architecture & Component Breakdown

```
cybercrime-predictive-framework/
├── backend/
│   ├── main.py                 # 16 REST routes, WebSocket broadcaster, state managers
│   ├── requirements.txt        # Python production dependencies
│   ├── data/
│   │   ├── atm_inventory.json      # 23+ geolocated ATMs with vulnerability metadata
│   │   └── indian_hotspots.json    # 8 curated crime corridors (coordinates, syndicates)
│   ├── models/
│   │   ├── schemas.py              # Pydantic models (Complaint, Prediction, ATM, Alerts)
│   │   └── predictive_engine.py    # Spatial KDE, Haversine, ETA regressor, confidence score
│   └── services/
│       ├── alert_dispatcher.py     # SMS, WhatsApp, and CFCFRMS Webhook emulator
│       ├── data_generator.py       # Synthetic NCRP/1930 fraud stream simulator
│       └── graph_forensics.py      # NetworkX DAG money-trail visualizer
├── database/
│   ├── schema.sql              # Production DDL for SQLite 3 / PostgreSQL
│   ├── cyber_netra.db          # Pre-seeded SQLite relational database
│   ├── seed_db.py              # Seeder (generates 100 complaints, 237 hops, 100 preds)
│   ├── DATA_DICTIONARY.md      # Comprehensive ERD and field definitions
│   └── datasets/               # 8 CSV/JSON dataset exports for open research
└── frontend/
    ├── src/
    │   ├── App.jsx                 # Central state manager, WebSocket listener, CSV export
    │   ├── main.jsx                # React 19 entry point
    │   ├── index.css               # Tailwind 4 theme, Indian Tricolor header animation
    │   ├── services/api.js         # API client & reconnecting WebSocket wrapper
    │   └── components/
    │       ├── Navbar.jsx              # MHA header, ticker bar, role switcher, export/sound tools
    │       ├── Auth/LoginPage.jsx          # Government login gate with sessionStorage isolation
    │       ├── Heatmap/TacticalGISMap.jsx   # Leaflet map, pulsing radar blips, time slider
    │       ├── Feed/PredictiveCashoutFeed.jsx # Live queue, search, batch freeze checkbox
    │       ├── Roles/RoleViews.jsx          # I4C Command, LEA Cell, Beat PCR, Bank Nodal desks
    │       ├── Graph/MoneyLayeringGraph.jsx # Interactive multi-hop DAG explorer
    │       ├── Alerts/AlertsDrawer.jsx      # Right drawer logging SMS, WhatsApp, API packets
    │       └── Modals/
    │           ├── DossierModal.jsx         # Section 91 CrPC / Section 94 BNSS notice generator
    │           ├── InjectIncidentModal.jsx  # Custom fraud injector with realistic presets
    │           └── HelpGuideModal.jsx       # 5-tab SOP, role guide, and legal handbook
```

## Complete API Route Matrix (`backend/main.py`)

| Method | Endpoint | Description | Role / Consumer |
|---|---|---|---|
| `GET` | `/api/status` | National daily metrics (complaints ingested, averted, Rs protected) | All roles / Navbar Ticker |
| `GET` | `/api/hotspots` | Geo-coordinates and syndicate density across Indian corridors | Tactical GIS Heatmap |
| `GET` | `/api/atms` | Complete inventory of ATMs with vulnerability & CCTV status | GIS Map / Target Selector |
| `GET` | `/api/complaints` | Recent NCRP 1930 fraud complaint records | District Cyber Cell |
| `GET` | `/api/predictions/active` | Live list of active predicted cash-outs with countdown timers | Predictive Cashout Feed |
| `GET` | `/api/graph/{complaint_i` | NetworkX nodes & edges for multi-hop money layering trail | Money Layering Modal |
| `POST` | `/api/dispatch` | Transmit instant GPS coordinates & suspect profile to PCR patrol | Beat Patrol / LEA Officer |
| `POST` | `/api/freeze` | Issue real-time total or debit-freeze lien to CFCFRMS bank API | Bank Nodal Officer |
| `POST` | `/api/simulate/inject` | Inject simulated cybercrime complaints with custom parameters | Test Scenario Simulator |
| `GET` | `/api/alerts` | Chronological log of generated SMS, WhatsApp, and API alerts | Alerts Drawer |
| `GET` | `/api/dossier/{id}` | Formatted statutory notice payload under Sec 94 BNSS / 91 CrPC | Judicial Evidence Modal |
| `WS` | `/ws` | Real-time bi-directional WebSocket streaming live threats | Control Room Dashboard |

## Database Entity Relationship Architecture (`database/schema.sql`)

The persistent layer comprises **8 normalized tables**. The core relationships are:

- `HOTSPOTS` contains many `ATMS`, and is the origin of many `COMPLAINTS`.
- `COMPLAINTS` generates many `LAYER_HOPS` and is forecasted into many `PREDICTIONS`.
- `PREDICTIONS` triggers `INTERVENTIONS` and dispatches `ALERT_LOGS`; each is targeted at an `ATM`.
- `MULE_SYNDICATES` operate within one or more `HOTSPOTS`.

**Key entity fields:**

| Entity | Field | Type |
|---|---|---|
| **HOTSPOTS** | `id` | string (PK) |
| | `name` | string |
| | `centroid_lat` | float |
| | `centroid_lon` | float |
| | `active_mule_density` | int |
| **ATMS** | `id` | string (PK) |
| | `name` | string |
| | `bank` | string |
| | `latitude` / `longitude` | float |
| | `atm_type` | string |
| | `vulnerability_score` | float |
| | `nearest_police_station` | string |
| **COMPLAINTS** | `id` | string (PK) |
| | `ack_id` | string |
| | `crime_category` | string |
| | `amount_stolen_inr` | float |
| | `complaint_timestamp` | timestamp |
| **PREDICTIONS** | `id` | string (PK) |
| | `complaint_id` | string (FK) |
| | `target_atm_id` | string (FK) |
| | `eta_minutes` | int |
| | `confidence_score` | float |
| | `risk_level` | string |
| | `intervention_status` | string |

## Role-Based Access Control (RBAC) & Governance Desks

CYBERSIGHT adapts its interface dynamically according to the logged-in official's role:

| 1. I4C Command | 2. District LEA | 3. Beat Officer | 4. Bank Nodal |
|---|---|---|---|
| National KPIs | Incident queue | Mobile layout | CFCFRMS queue |
| Interstate nexi | Officer dispatch | Turn-by-turn GPS | 1-Click lien |
| State rankings | Case dossiers | ATM CCTV photos | Account freeze |
| Policy view | Graph forensics | Suspect profile | Batch freezes |

## Statutory Legal Framework Grounding

CYBERSIGHT is strictly aligned with the statutory procedural laws of India:

1. **Section 94, Bharatiya Nagarik Suraksha Sanhita (BNSS 2023)** *(formerly Section 91 CrPC)*: Empowers police officers to demand the production of documents or electronic records. The platform auto-generates legal notices directly summoning bank records and ATM footage within 2 hours.

2. **Section 66 & 66D, Information Technology Act 2000**: Covers identity theft and cheating by personation using computer resources.
3. **Reserve Bank of India (RBI) Master Directions on Cyber Fraud**: Powers the instantaneous placement of a "Total Freeze" or "Debit Lien" on verified Layer-1 and Layer-2 mule accounts under the CFCFRMS 1930 framework.

## Codebase Knowledge Graph Audit (Powered by `graphify`)

An automated structural audit was executed across all source files in the project. The results confirm architectural integrity:

- **Total Nodes:** 151
- **Total Directed Edges:** 301
- **Detected Communities:** 9 distinct operational clusters
- **Import Cycles:** 0 (Zero circular dependencies)

**Dominant Core Abstractions (God Nodes):**

| Rank | Node | Edges | Role |
|---|---|---|---|
| 1 | `Complaint` | 13 | Central data bus connecting all pipelines |
| 2 | `react` | 13 | Cross-community bridge linking components |
| 3 | `PredictiveAnalyticsEngi` | 12 | Mathematical intelligence nucleus |
| 4 | `RiskLevel` | 12 | Shared threat evaluation enumeration |
| 5 | `Prediction` | 12 | Actionable intelligence carrier |

## Verification & Operational Demonstration

CYBERSIGHT includes full simulation capabilities:

- **Interactive Scenario Injector:** Officials can test hypothetical fraud complaints (e.g., a Rs 4,50,000 Digital Arrest scam originating in Mewat), instantly visualizing how the AI models the trail, ranks ATMs, updates the GIS radar, and generates dispatch packets.
- **Control-Room Audio Signals:** Utilizes native Web Audio API oscillators to provide sound cues during live command center monitoring without external media assets.
- **Auditable CSV Reports:** 1-Click downloads of forensic data for senior leadership briefings and court charge sheets.

*Authored for official submission and technical evaluation under Smart India Hackathon 2026.*